

Algorithmic Bias

Systematic and unfair discrimination in the outputs of an algorithmic system

GLOSSARY 06 - SHEET 3 / 8

PROFESSIONAL DEFINITION

Algorithmic bias is systematic and repeatable error in the output of a computational system that creates unfair outcomes, such as privileging one group of users over another. It arises from pre-existing bias embedded in the training data or in social practice, technical bias arising from the design of the system, and emergent bias arising when a system is used in a context different from the one for which it was built (Friedman and Nissenbaum, 1996; Mehrabi et al., 2021).

WHO DEFINED IT ORIGINALLY

Batya Friedman and Helen Nissenbaum established the three-part taxonomy in ACM Transactions on Information Systems in 1996. The contemporary empirical literature is anchored by Joy Buolamwini and Timnit Gebru's (2018) Gender Shades audit of commercial facial-analysis systems; Ninareh Mehrabi and colleagues (2021) provide the systematic survey of sources and mitigations.

WHY THIS DEFINITION WAS DEVELOPED

Automated systems were presented as neutral because they were mechanical, which obscured the fact that they encode the distribution of the data used to build them and the assumptions of those who built them. Friedman and Nissenbaum needed a taxonomy that distinguished sources, because each source requires a different remedy and the same observed disparity may have any of the three.

FIGURE 1 - THREE SOURCES OF BIAS

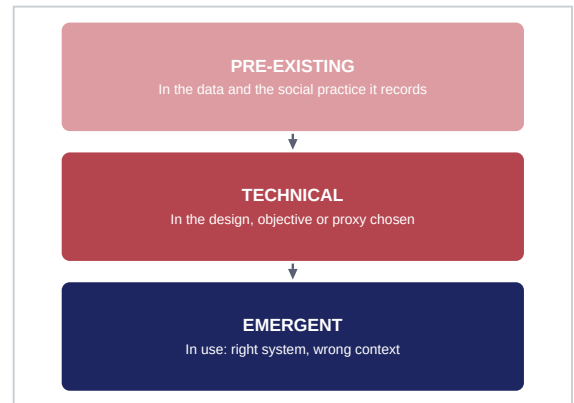


Figure 1. The three sources require different remedies and can produce identical observed disparities. Rebalancing the training data addresses only the first, which is why audits that stop at data composition frequently fail to remove the disparity (after Friedman and Nissenbaum, 1996).

HOW IT IS USED

Audit outputs by subgroup rather than in aggregate, since aggregate accuracy can be high while subgroup accuracy is not; then identify which of the three sources is operating, because data rebalancing does not fix a technical or emergent bias.

WHEN IT IS USED

Before deployment of any system affecting individuals, at periodic audit of a live system, when a subgroup disparity is reported, and whenever a system is used in a new population or context.

BY WHOM IT IS USED

Model validation and audit functions, data scientists selecting training data, legal and compliance teams assessing discrimination exposure, and deployers of vendor systems built on unknown data.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The predecessor treatment framed bias as a statistical property of an estimator, which addressed variance from a true value but not fairness across groups. The Friedman and Nissenbaum taxonomy relocates the question to sources and to social consequence, and identifies emergent bias - correct system, wrong context - which purely statistical framings cannot express. Buolamwini and Gebru's disaggregated audit method then converted the argument into a measurement protocol that could be run on a commercial system from the outside.

RELATED DEFINITIONS IN THIS COURSE

[AI Ethics - Unit 6](#)
[Explainability - Unit 6](#)
[AI Governance - Unit 6](#)
[Digital Divide - Unit 5](#)
[Trustworthy AI - Unit 6](#)

SOURCES (HARVARD REFERENCING STYLE)

- Buolamwini, J. and Gebru, T. (2018) 'Gender shades: intersectional accuracy disparities in commercial gender classification', Proceedings of Machine Learning Research, 81, pp. 77-91.
- Friedman, B. and Nissenbaum, H. (1996) 'Bias in computer systems', ACM Transactions on Information Systems, 14(3), pp. 330-347.
- Mehrabi, N., Morstatter, F., Saxena, N., Lerman, K. and Galstyan, A. (2021) 'A survey on bias and fairness in machine learning', ACM Computing Surveys, 54(6), pp. 1-35.